Fiscal Year:	FY 2009	**Task Last Updated:**	FY 02/27/2009
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Crop and Bio Mass Production		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2009 per PI (1/09) NOTE: Received NCE to 11/30/2008 per PI (11/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Beyl, Caula (Alabama A & M) Mitchell, Cary (Purdue University) Brown, Paul (Purdue University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Three NSCORT (NASA Specialized Center of Research and Training) projects address the feasibility of producing edible biomass in a space regenerative life-support system. One project addresses a novel approach to significantly reduce the ESM (Equivalent System Mass) of growing food crops in controlled environments; the second project tests the potential to use fungi to degrade non-edible crop residues while generating edible mushrooms; and a third project addresses the feasibility of feeding crop or bioreactor residue to fish that can be eaten. A novel lighting system for higher plants has demonstrated incremental improvement in electrical energy utilization during the project by correcting software glitches, by creating growth compartments that reflect escaped photons back into the crop stand, by planting dense crop stands that close fast, by positioning plants directly in front of one-sided light strips, and by relying upon the "plasticity" of plant growth to seek out light. Appropriate red/blue ratios to maintain this "light magnet" are being defined. Intracanopy lighting is superior to overhead lighting for crop quantum efficiency and electrical-energy utilization once foliar canopies have closed. It has been confirmed that reproductive development occurs normally with intracanopy LED lighting. Remote position sensing of canopy height and automated switching of light engines is in development. Once needed engineering upgrades to the prototype intracanopy lighting system are implemented, ESM for crop lighting should be reduced at least an order of magnitude relative to that of traditional overhead lighting systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. Development of solid-state, intracanopy and close-canopy crop-lighting systems will solve the light-distribution problem that has long limited the photosynthetic productivity of overhead-lighted crop stands, and will permit the use of low-power light sources. These advances will enable year-round, warehouse-based, controlled-environment agriculture (CEA) production of salad greens, herbs, and certain vegetables to be profitable in local markets, especially if coupled with sources of cheap, bio-derived energy, and competitive with produced trucked from distant production sites. 2. Developing methods for maximum crop yield at minimum cost in controlled-environment agriculture (CEA) 3. Developing genetic engineering of CEA crops for superior performance
Task Progress:	ALS Candidate Crop Cultivation: Strawberry. Strawberry is an appealing small fruit to provide nutritional and psychological support for space-crew diets. In addition to having positive organoleptic qualities (i.e., flavors, aromas, texture), it also has been found to contain compounds that protect brain function against ionizing radiation of the type that exists in space. The Mitchell Lab has been surveying strawberry cultivars for ability to produce highly palatable fruit consistently over a long period of time. Strawberry cultivars have been surveyed under a number of different optimizing thermal and light environments and cultural conditions. The cultivar ‘Seascape’ has been identified as a promising day-neutral strawberry that can flower and produce fruit year-round. Environments need to be optimized for ‘Seascape’ to bring out the best flavor and to save maximal electrical energy for crop lighting by growing it under the shortest possible days that do not limit fruit yield or quality. Cultural practices that need optimization include semi-automation of pollination and discouragement of vegetative runner formation in favor of flowering and fruiting. Taste-panel evaluations as well as productivity and yield optimizations will be performed during the coming year. Rejuvenation approaches such as cutting back or separation will be compared with new seed stock to maintain plants in a productive condition for crews at a future space base. ALS Candidate Crop Cultivation: Sweetpotato. Sweetpotato not only produces large amounts of carbohydrate for space-crew diets in its tuberous roots, but considerable anti-oxidants and vitamins as well in both roots and young, edible shoots, which will be important for protection against deleterious physiological effects of space radiation. Sweetpotato shoots typically grow aggressively and occupy too much growing area to be considered for use in limited growth spaces. The Mitchell Lab has been using various approaches to contain sweetpotato vines without compromising root yield. Helical winding of shoots around wire supports coupled with removal of edible lateral buds has successfully contained inedible mature vines within acceptable growth areas and volumes while maintaining productive root yield. During the coming year, water and fertilizer resources will be optimized by limiting nitrogen availability for shoot growth while maintaining phosphorus, potassium, and micronutrient levels to encourage storage root production. Solid substrate materials will be surveyed for root yield and potential for reusability at a space base. Irrigation frequency and amount will be varied to determine minimum water needs and to maximize water-use efficiency so that Equivalent System Mass for sweetpotato production will keep this valuable food crop on the menu for certain life-support scenarios. LED Lighting Project: Light-Quality Optimization. The cool light-emitting surfaces, ruggedness, and long life times of light-emitting diodes strongly suggest LEDs as the preferred electric-lighting source for space crops in the future. The Mitchell Lab has investigated novel ways to save energy in delivering light to crops in controlled environments, including the use of vertical “lightsicles” that hang within crop stands and around which crops grow, thereby delivering light to all photosynthetic tissues within a foliar canopy. Because LEDs come in narrow spectra that can be selected to match absorption spectra of leaves, plants are grown using mainly red and blue LEDs that provide the wavelengths absorbed most efficiently by chlorophyll. Because red LEDs are more efficient in converting electricity into light than are blue LEDs, during the coming year the Mitchell Lab will determine how much additional lighting energy can be saved by minimizing the blue-light contribution to crop growth. Variables such as proportion of blue, duration of blue, and time of day when blue is used will be tested for several ALS candidate crop species known to have a requirement for blue light. LED Lighting Project: Canopy Gas Exchange. A combination controlled environment chamber / crop-stand gas-exchange cuvette has been under design and construction for the past two years and is in the assembly phase within the Mitchell Laboratory. During the coming year, this valuable tool will be used to rapidly determine optimizing conditions for light level and spectral composition in the intracanopy light-distribution mode, as well as CO2 concentration for multi-plant stands of ALS candidate crop species for space life support. Photosynthetic CO2 uptake rates will be used as a real-time metric of combined environmental conditions that can be varied and re-equilibrated quickly to determine maximum output (photosynthesis) for minimum inputs (especially electrical energy for lighting). Combined with the energy-saving assets of LEDs and the intracanopy lighting configuration, the optimizing crop-production environments to be defined by gas-exchange rates suggest order-of-magnitude savings in electrical energy for growing crops for life support in space. Upcoming Productivity Metrics of the Edible Biomass / Crop-Production Group. During rthe final year of the ALS NSCORT, the Mitchell Lab will publish peer-reviewed articles on strawberry, sweetpotato, crop requirements for

	intracanopy vs. overhead red / blue LED lighting, and performance demonstration of the crop cuvette system. Dr. Gioia D. Massa, a highly experienced, well-published post-doctoral research associate, will become available for employment opportunities in the space life science sector.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Articles in Peer-reviewed Journals	Massa G, Kim HH, Wheeler R, Mitchell C. "Plant productivity in response to LED lighting." HortScience, 2008 Dec;43(7):1951-6. http://hortsci.ashspublications.org/cgi/content/abstract/43/7/1951 , Dec-2008
Articles in Peer-reviewed Journals	Gonzales JM, Hutson AH, Rosinski ME, Wu YV, Powless TF, Brown PB. "Evaluation of fish meal-free diets for first feeding Nile tilapia, Oreochromis niloticus." Journal of Applied Aquaculture, 2007 Aug;19(3):89–99. http://dx.doi.org/10.1300/J028v19n03_06 , Aug-2007